

QuEra challenge for ETH Quantum Hackathon 2026:

Can a 1-Qubit gate be harder than a 2-Qubit gate?

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The greatest deception in quantum computing is the simplicity of a single line on a circuit diagram. On paper, a 1-qubit rotation like $R_z(\pi/2^k)$ is a triviality—a mere twist of a vector on the Bloch sphere. But as we transition from the "toy model" of isolated atoms to the rugged reality of fault-tolerant architectures, we encounter a staggering paradox: a 1-qubit gate can be significantly harder to implement than a 2-qubit gate. This project is a deep dive into that friction, bridging the gap between the surgeon analysis of Team Synthesis and the architectural ingenuity of Team STAR.

We will begin this journey with a humble, exhaustive list of Clifford and T gate combinations—a brute-force search for the right sequence—only to realize that in a world governed by the Eastin-Knill Theorem, not all gates are created equal. While 2-qubit gates like the $CNOT$ are often "transversal" (meaning they can be applied bit-wise across logical blocks without spreading errors), the non-Clifford T gate is an elusive rebel. It cannot be applied directly to a $[[7,1,3]]$ Steane logical qubit without compromising its integrity. This forces us into a world of "Magic State Injection" and "Teleportation," where a single rotation is no longer a pulse, but a complex choreography of auxiliary qubits and post-selection.

It is at this point where the work of our two teams converges: Team Synthesis explores the optimal "recipe" for building these rotations using the fewest expensive T gates possible, while Team STAR builds the "hardware gadgets" necessary to inject these rotations into the logical flow. By analyzing the move from a single physical qubit to a protected logical qubit, we expose the true "tax" of quantum error correction. We are no longer just approximating an angle; we are benchmarking the massive overhead in circuit depth and qubit count required to keep a quantum calculation from collapsing. This project isn't just about gates—it's about understanding the high price of precision in the fault-tolerant era, where "simple" 1-qubit rotations become the ultimate test of a quantum processor's scalability.

1 Synthesis

Our journey through the Team Synthesis track began with the raw, computational challenge of approximating the $R_z(\pi/2^k)$ family of rotations using the discrete Clifford+T gate set. The initial

strategy was a classic brute-force first search. We treated every possible sequence as a path in a binary tree, where each node branched into an application of either an H or a T gate. However, we quickly encountered the brutal reality of exponential scaling. In quantum circuit synthesis, the number of possible sequences grows as 2^n . By the time we pushed our search depth to $n = 25$, the state space had ballooned to over 33 million possible combinations. This created a significant "computational wall," where a single search took upwards of 10 minutes, making it impossible to iterate quickly or reach the high-fidelity approximations needed for deeper rotations.

To dismantle this wall, we first applied a layer of algebraic pruning to the search tree. We utilized the fundamental identities of the gate set, specifically $H^2 = I$ and $T^8 = I$, to automatically discard redundant paths. By enforcing these rules at the compiler level, we prevented the algorithm from exploring thousands of dead-end branches. This sharpened our focus on sequences that actually progressed the state vector across the Bloch sphere, but it wasn't enough to solve the 2^n scaling issue for the very small angles required in Part 2.

The "eureka" moment for our optimization came through a technique we call Operator Tokenization. We noticed that the most successful approximations weren't just random strings of gates; they were composed of recurring, high-utility blocks such as HT , $HTHT$, TT , or $TTTT$. Instead of building circuits gate-by-gate, we refactored our search to use these blocks as "macro-operators". This fundamentally altered the math of our search: we shifted from a 2^n complexity (where n is a large depth) to a k^m complexity (where k is a slightly larger branching factor of 5 or 6, but m is a significantly smaller number of "steps"). This allowed us to leapfrog through the Hilbert space, reaching effective depths that would have taken days to calculate using the original gate-by-gate method.

Using this optimized engine, we were able to achieve distances of approximately 0.03 almost instantaneously. In the context of the distance metric provided

$$d(U, V) = \sqrt{1 - |\text{Tr}(U^\dagger V)|^2/2} \quad (1)$$

this value represents an incredibly tight fit. In Figure 1 the value of this distance is compared for three cases, for growing H-T chain length. To put this in perspective, for tiny rotations like $\pi/128$, this error is an order of magnitude better than a "null" rotation (simply doing nothing) for just 20 gate-deep approximation. This level of precision is critical because, as the target angle $\pi/2^k$ becomes smaller, the margin for error shrinks. Our strategy ensured that we weren't just guessing; we were systematically finding the most efficient "path" on the sphere to reach our destination with minimal resources.

This leads us to the transition into Part 3, where the rules of the game changed. We had to account for the fact that in a fault-tolerant setting, T gates are "expensive" and cannot be applied directly to the data qubit due to their non-transversal nature. We took our optimized sequences and reimagined them as a series of T-gate injections. This required a more complex architecture (shown in Figure 2): an auxiliary qubit is prepared in a "magic state" $|H\rangle$, which essentially acts as a pre-packaged rotation. By entangling this auxiliary qubit with our main qubit through a $CNOT$ and performing a projective measurement, we "teleport" the T rotation onto the main qubit.

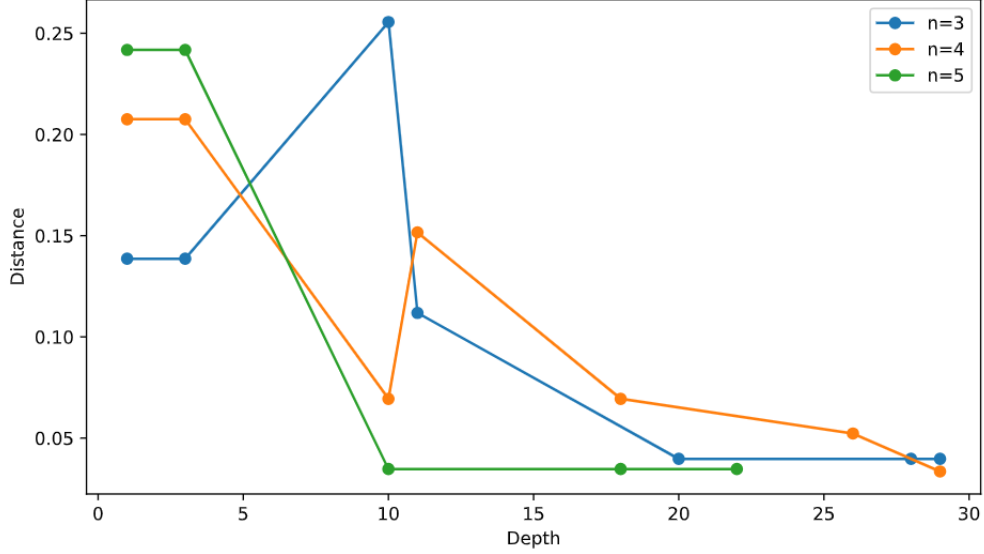


Figure 1: A considerable reduction in distance is observed for higher depths of the H and T chain, for all $\pi/8$, $\pi/16$ and $\pi/32$.

The final stage of our strategy involved managing the "tax" of this injection. Because the measurement result is probabilistic, we implemented a feed-forward correction system. If the measurement yields a '1', the qubit ends up with an unwanted phase that must be corrected using a controlled- S gate. This transition from a single-qubit "pulse" to a multi-qubit choreography of entanglement, measurement, and classical feedback is the core of our project. It proves that while a rotation might look like a single line in a textbook, implementing it logically requires a sophisticated marriage of synthesis optimization and hardware-aware circuit design. This synergy is exactly what allows us to bridge the gap between a physical qubit and a protected, logical Steane qubit.

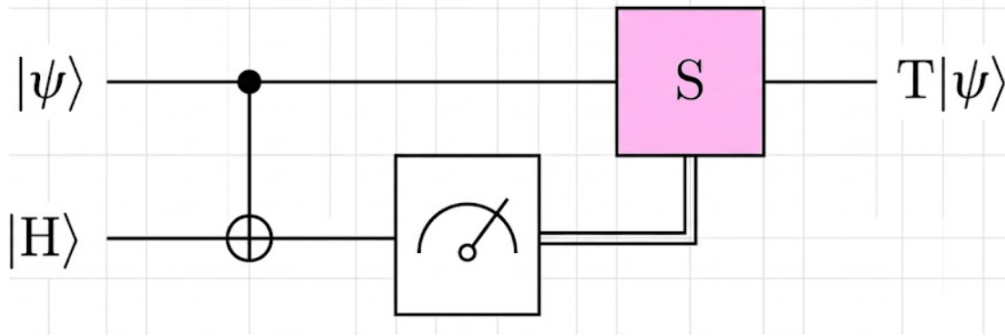


Figure 2: T-gate injector routine scheme.

To test the accuracy attained by means of the T-gate injection methods, we carried out a random sampling of different states and checked the matching between the direct T-gate application and the injected one. In Figure 3, we offer the comparison of the 0 and 1 components of both. Probabilities were inferred from law of large number (experiment repetition).

When we transition from a single physical qubit to the $[[7,1,3]]$ Steane code, we enter the realm of Fault-Tolerant Quantum Computing (FTQC). The Steane code is a CSS (Calderbank-Shor-Steane)

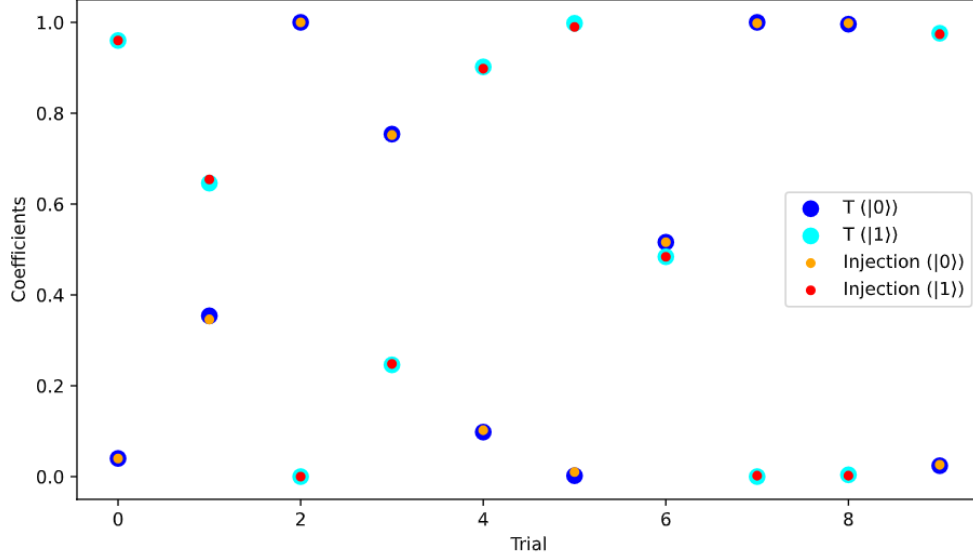


Figure 3: Comparison of 0 and 1 components for 10 different random samples.

code, which is particularly elegant because it allows all Clifford gates—like H , S , and $CNOT$ —to be implemented transversally. To apply a logical Hadamard H_L , you simply apply a physical H to each of the seven qubits in the block. It’s clean, it’s symmetrical, and it doesn’t spread errors between qubits.

However, we immediately run into a fundamental wall known as the Eastin-Knill Theorem. This theorem states that no quantum error-correcting code can implement a universal set of gates using only transversal operations. For the Steane code, the “rebel” in the set is the T gate. If you were to apply a physical T gate to each of the seven qubits, you wouldn’t end up with a logical T gate; instead, you would rotate the state out of the code space, effectively breaking the very protection we worked so hard to build.

To implement a logical T gate, we have to use a technique called Magic State Injection. Since we cannot apply the T rotation directly to our data, we prepare a separate “resource” block in a very specific state called the Logical Magic State, denoted as $|T\rangle_L$. This state is mathematically defined as:

$$|T\rangle_L = \frac{1}{\sqrt{2}}(|0\rangle_L + e^{i\pi/4}|1\rangle_L)$$

Preparing this state is a task in itself—it often requires a process called distillation to ensure the “magic” is pure and free of noise. Once we have the resource used in Part 3, we interact our data block $|\psi\rangle_L$ with the magic state block $|T\rangle_L$ using a logical $CNOT$. Because the $CNOT$ is transversal in the Steane code, this interaction is “safe”.

After the entanglement, we perform a logical measurement on the auxiliary block. This is where the “injection” happens: the measurement collapses the entanglement and “teleports” the $\pi/4$ phase shift onto our main data qubit. Since the measurement is probabilistic, half the time we get the exact T rotation we want. The other half of the time, we accidentally apply a $T^\dagger$ rotation. To fix this, we use a feed-forward correction: if the measurement result is ‘1’, we apply a logical S gate (which is transversal and “cheap”) to rotate the state back to the correct target.

This process reveals the true "cost" of logical quantum computing. To perform a single T gate that we synthesized in Part 2, we now require:

- A minimum of 14 physical qubits (7 for data, 7 for the magic state).
- 24 gate-operations for each logical qubit preparation (48 in total), plus the 3 gates for applying T_L to ψ_L .
- Real-time classical feedback to decide whether to apply the S correction.

When you look at the sequences we generated earlier—long strings of H and T gates—the overhead becomes staggering. A sequence of ten T gates isn't just ten pulses; it's ten separate magic state injections, each requiring its own auxiliary block and measurement cycle. This is precisely why our synthesis strategy from Part 2 is so vital: every T gate we can eliminate through clever optimization isn't just saving a pulse—it's saving an entire architectural "gadget" in the Steane code. This transition from "gate-pushing" to "resource-management" is the defining challenge of the fault-tolerant era.

2 The Illusion of Simplicity in Fault-Tolerant Quantum Gates

The apparent simplicity of a gate at the physical level is misleading once information is encoded fault-tolerantly. A continuous single-qubit rotation may be implemented on a bare qubit by tuning a control pulse, yet this intuition does not survive at the logical level. In an error-corrected architecture, the most natural operations are the transversal Clifford gates, whose action preserves the stabilizer structure of the code. For this reason, gates such as the logical $CNOT$ are comparatively benign: they respect the protected code space and can be monitored by standard syndrome extraction without converting local faults into catastrophic correlated errors.

The difficulty of logical one-qubit rotations emerges from the Eastin–Knill theorem, which forbids any quantum error-correcting code from implementing a continuous, universal set of gates transversally. The consequence is structural rather than merely technical: the non-Clifford rotations required for universality cannot be implemented in a way that simultaneously remains continuous, transversal, and fully compatible with the stabilizer protection of the code. Geometrically, a continuous rotation drags the state away from the invariant stabilizer subspace and into the analog bulk of the Hilbert space. Because the error-correcting stabilizers are explicitly designed to project analog drift back into discrete Pauli errors, the code itself will actively collapse and destroy the intended continuous rotation during syndrome measurement. Thus, the true challenge of a logical one-qubit rotation is performing a continuous operation without triggering the discrete protection mechanisms that make fault tolerance possible.

This obstruction explains the central role of resource-intensive alternatives such as magic-state distillation, in which the desired non-Clifford operation is prepared indirectly and injected into the computation through teleportation. The STAR (Space-Time-efficient Analog Rotations) paradigm explores a different route: it realizes the target logical action through a carefully calibrated analog rotation applied across a transversal boundary of the code. Although elegant in concept, this

strategy comes at a cost. Because the analog evolution does not commute with the stabilizer measurements in the same way as a Clifford operation, standard active correction can no longer be assumed to protect the process throughout the gate.

As a result, the protocol shifts from error correction to error detection. The logical rotation is executed on an ancillary encoded state, and the computation is accepted only if the surrounding syndrome checks remain silent. Any detector event invalidates the run and forces its rejection. This post-selection mechanism is precisely where the hidden expense of the “simple” one-qubit gate becomes visible: increasing the code distance improves baseline protection, but it also enlarges the physical support of the analog operation and thereby suppresses the probability of a clean, accepted shot.

Classical simulation makes this trade-off quantitative. To evaluate the protocol faithfully, one must relate the physical rotation angle θ_{phys} to the target logical angle θ_{logical} , propagate the resulting coherent error through the encoded circuit, and condition the analysis on successful post-selection. In practice, this means that meaningful estimates of the logical error rate require a very large number of samples, since most shots are discarded before contributing to the final statistic. The numerical conclusion mirrors the architectural one: in the fault-tolerant regime, the dominant cost of a logical one-qubit rotation lies not in generating the angle, but in preserving the encoded computation while that angle is applied.

2.1 Hacking our way through Surface coding (what we did)

We started getting familiar with the quantum error-correction scheme of the surface code. Specifically, we focused on the distance-3 surface code, with 9 data qubits and 8 ancilla qubits. We learned how to arrange them so that the stabilizers could reveal any bit-flip or phase-flip error affecting a physical qubit, and how these errors propagate through the circuit and are detected by comparing the ancilla-qubit measurement outcomes, a process known as syndrome extraction.

We created the circuit for detecting and correcting these simple errors successfully. See the circuit below.

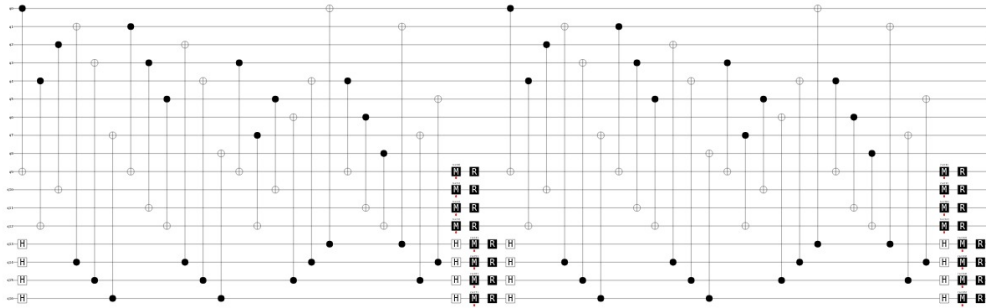


Figure 4: Distance 3 surface code written to see how errors are detected.

We also understood how 17 physical qubits encode a single logical qubit in a fault-tolerant way.

Once we understood this, we moved to learning how to read, visualize, and test already-built STAR (Space-Time Efficient Analog Rotations) circuits. We studied how these circuits propose a different route for implementing non-Clifford gates in a fault-tolerant setting, avoiding the usual resource overhead of the standard approach.

Instead of focusing on magic-state distillation, we analyzed the STAR circuits directly. We visualized them and understood their structure: in a noisy circuit, the protocol first initializes a logical $|\bar{+}\rangle$ state using standard digital syndrome extraction, locking the state into the protected code space. The key operation occurs midway through the protocol, where a continuous non-Clifford analog rotation is applied simultaneously to a connected string of d physical qubits stretching across the logical patch. In this way, the physical analog evolution induces the desired logical rotation on the encoded qubit.

However, this architectural efficiency comes at a strict operational cost. Because the analog rotation does not behave like the usual stabilizer-preserving operations, the subsequent syndrome extraction rounds act mainly as an error-detecting gadget rather than a fully error-correcting one. If any parity check detects a localized physical fault, the corresponding detector fires, and the classical controller must post-select and discard that shot.

We then tried to replicate one of the plots from the instructions by studying how the distance-normalized logical error rate of the STAR architecture scales with the applied rotation angle across distances $d = 3, 5, 7$.

Because Tsim is strictly a Clifford simulator, we first employed a Pauli approximation, mathematically mapping the non-Clifford transversal analog rotations ($R_z(\theta)$) to probabilistic Z -error channels, allowing us to natively simulate the coherent noise injected by the protocol at scale. We modified the `stim` files to run the STAR circuits over a chosen range of physical and logical rotation angles. During execution of the `stim` file, we extracted the raw classical records and separated the syndrome detectors from the final logical observables. Instead of naively post-selecting the entire circuit, we applied strict post-selection exclusively to the $3(d^2 - 1)$ detectors active during the transversal rotation phase, discarding any shot where the analog gate triggered a localized error. For the surviving computational shots, we fed the remaining stabilizer measurements into a Minimum Weight Perfect Matching (MWPM) decoder to actively predict and correct errors occurring during the standard digital memory phase. Finally, we compared these MWPM predictions against the empirical logical measurements. By simulating up to 25 million shots per data point to overcome the severe shot-noise limit imposed by the post-selection bottleneck, we accurately calculated the distance-normalized logical error rates, clearly resolving the trade-off between topological protection and transversal analog noise.

From the plot we observe that, for small rotation angles, the logical errors were dominated mainly by circuit noise, with lower error rates as the code distance increases. However, we also observed that, for all three distances, the logical error rate started to increase in a linear way, reaching higher and equal rates as the rotation angle becomes larger. This reflects the central trade-off of the STAR protocol: implementing the logical gate through a transversal analog rotation requires acting coherently on a string of d data qubits at once, and at large angles this induces coherent logical errors that the surface-code protection cannot fully suppress.

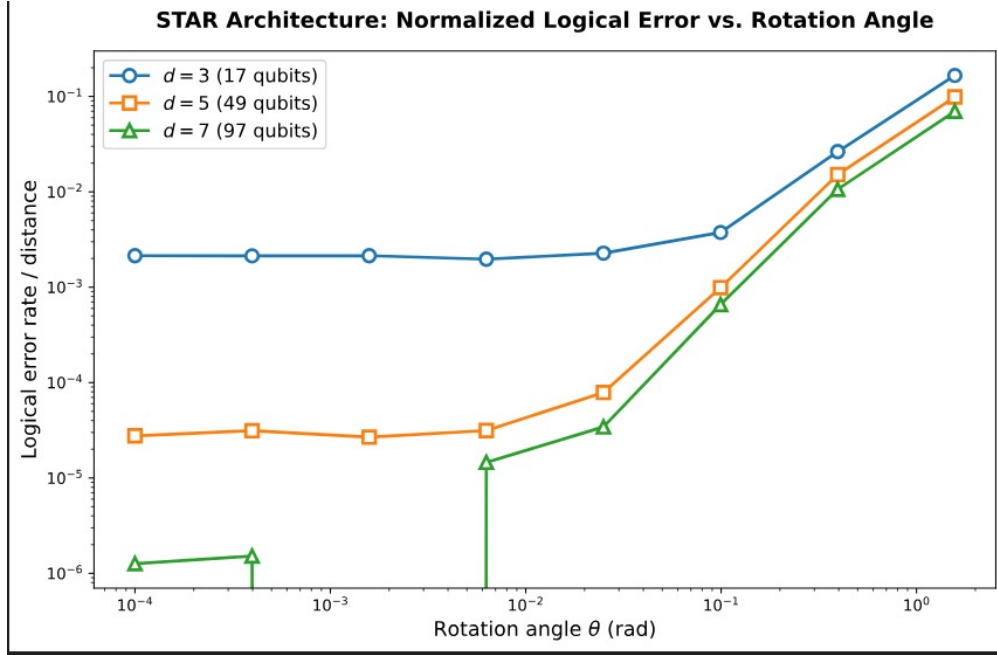


Figure 5: Simulation of distance-normalized logic qubit error rate as a function of the rotation angles.

3 Use of AI

We used AI to try to make our lives easier. We used it...

- to help us guide us and understand certain concepts in a fast way, contrasting everything with papers and other sources in the internet
- to review written text and code, specially to debug typical annoying programming bugs in the code that are not important or relevant for a physicist.